

Statistics

2025 Lectures

Part 6 - Selected Families of Distributions

Institute of Economic Studies
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Recommended reading

For this lecture Part 6, please, read

- sections 8.1, 8.2, 8.3, 8.4, 8.5 from Bartoszynski and Niewiadomska-Bugaj (42 pages)
- sections 4.1, 4.2, 4.3, 4.4 from Mittelhammer

Bernoulli trials and Alternative (0-1) distribution

- **Bernoulli trials** are independent repetitions of an experiment in which we are interested in a particular event A (a “success”) that occurs in each trial with the same probability p , while the complementary event A^c (a “failure”) occurs with probability $1 - p$.
- $X = \begin{cases} 1 & \text{if } A \text{ occurs with } P(X = 1) = p \\ 0 & \text{if } A^c \text{ occurs with } P(X = 0) = 1 - p = q \end{cases}$
- we write $X \sim ALT(p)$
- $E(X) = p$ and $Var X = pq$
- since $0^n = 0$ and $1^n = 1$,

$$m_n = E(X^n) = E(X) = p, \quad \forall n \geq 1$$
$$m_X(t) = E(e^{tX}) = e^{t \cdot 0}q + e^{t \cdot 1}p = pe^t + q, \quad t \in \mathbb{R}.$$

Binomial distribution

- **Binomial distribution** plays a central role in probability theory and statistics, referring to the total number of successes in n Bernoulli trials.
- $S_n = X_1 + \dots + X_n$, where $X_i \sim ALT(p)$, $i = 1, \dots, n$ are iid
- We write $S_n \sim BIN(n, p)$

$$P(S_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n.$$

- $E(S_n) = np$ and $Var S_n = npq$

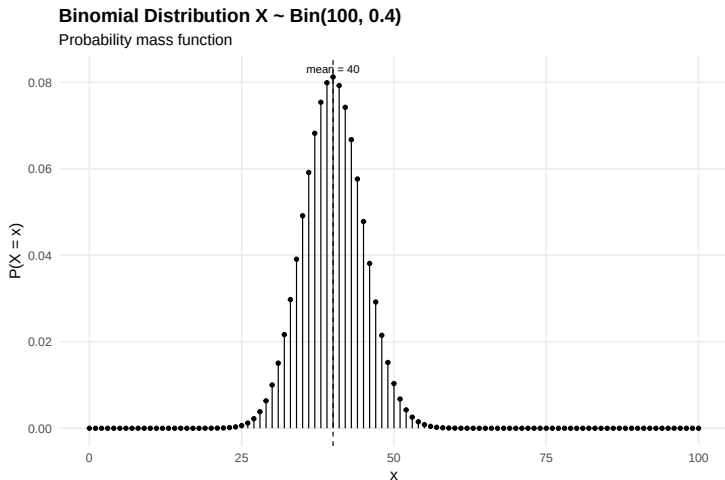
$$m_{S_n}(t) = (pe^t + q)^n, \quad t \in \mathbb{R}.$$

- If $S_n \sim BIN(n, p)$ then $S'_n = n - S_n \sim BIN(n, 1 - p)$ useful for cases when the role of a “success” and of a “failure” is reversed.

Binomial distribution: Example

Example 61: Consider $X \sim \text{BIN}(100, 0.4)$.

The graph of the **probability mass function**



Binomial distribution

Example 62: Assume that approximately 1 in 80 births results in twins. What is the probability that there will be no twins births among the next 200 births in the maternity ward of a particular hospital?

Let $S_{200} \sim \text{BIN}(200, 1/80)$ denotes the number of twin births among the next 200 births, we need to compute $P(S_{200} = 0)$.

$$\begin{aligned} P(S_{200} = 0) &= \binom{200}{0} \left(\frac{1}{80}\right)^0 \left(\frac{79}{80}\right)^{200} \approx 0.0808 \\ &= \left(1 - \frac{200/80}{200}\right)^{200} \approx e^{-200/80} \approx 0.0821. \end{aligned}$$

The exponential approximation is based on

$$\lim_{n \rightarrow \infty} \left(1 - \frac{c}{n}\right)^n = e^{-c}$$

which works well if the probability of a success p is small and the number of trials n is large.

Geometric distribution

- **Geometric distribution** refers to the number of failures that occur before the first success.
- $P(X = k) = q^k p$, $k = 0, 1, 2, \dots$ and we write $X \sim GEO(p)$
- geometric because $P(X \geq k) = q^k p + q^{k+1} p + \dots = q^k$
- also the distribution of rv $Y = X + 1$ (number of trials up to and including the first success) is called geometric

■

$$F_X(x) = \begin{cases} 0, & x < 0; \\ \sum_{0 \leq k \leq x} q^k p, & 0 \leq x < \infty. \end{cases}$$

- $E(X) = \frac{q}{p}$ and $Var X = \frac{q}{p^2}$

■

$$m_X(t) = \frac{p}{1 - qe^t} \text{ for } qe^t < 1 \text{ (else it is } +\infty \text{)}.$$

Theorem 33: (memoryless property of $GEO(p)$)

Geometric distribution has the memoryless property, i.e., for $X \sim GEO(p)$ and any $n, m \in \mathbb{N} \cup \{0\}$, one has

$$P(X \geq m + n | X \geq m) = P(X \geq n).$$

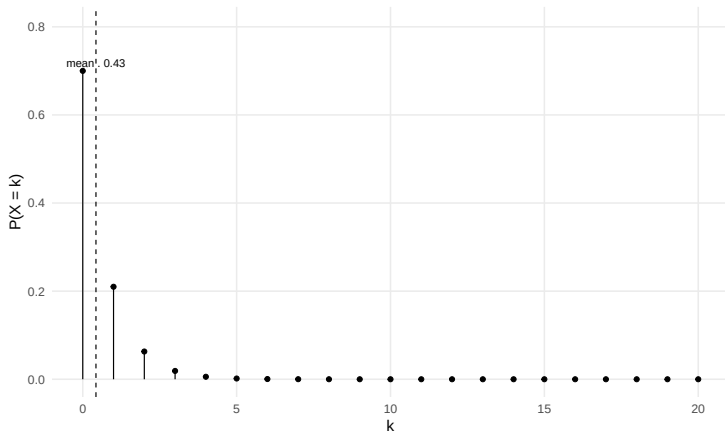
Geometric distribution: Example

Example 63: Consider $X \sim \text{GEO}(0.7)$.

The graph of the **probability mass function**

Probability mass function of Geometric($p = 0.7$)

X = number of failures before first success



Negative binomial distribution

- **Negative binomial distribution** refers to the number of Bernoulli trials required to achieve the r th success, inclusive.
- the event $\{Y = n\}$ occurs if the n th trial is a success and the first $n - 1$ trials give exactly $r - 1$ successes. Thus

$$P(Y = n) = \binom{n-1}{r-1} p^r q^{n-r}, \quad n = r, r+1, \dots$$

and we write $Y \sim NB(r, p)$.

- also rv $X = Y - r$, the number of failures preceding the r th success, is referred to as having a negative binomial distribution
- $E(Y) = \frac{r}{p}$ and $Var Y = \frac{rq}{p^2}$



$$m_Y(t) = \frac{p^r e^{tr}}{(1 - qe^t)^r} \quad \text{for } qe^t < 1 \text{ (else it is } +\infty \text{)}.$$

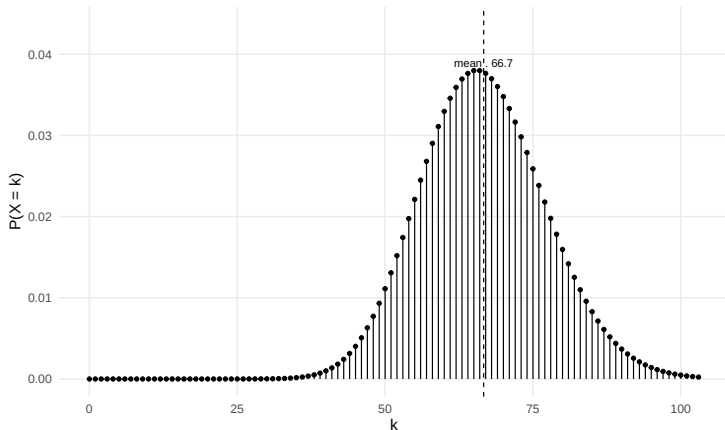
Negative distribution: Example

Example 64: Consider $X \sim NB(100, 0.6)$.

The graph of the **probability mass function** for Y which counts number of failures before a target number of successes is reached

PMF of Negative Binomial($r = 100$, $p = 0.6$)

X = number of failures before the 100th success



Negative binomial distribution

Theorem 34: Let $S_n \sim \text{BIN}(n, p)$ and $Y_r \sim \text{NB}(r, p)$. Then for every $k = 0, 1, \dots$

$$P(Y_r > k) = P(S_k < r).$$

Example 65: A salesman makes calls to prospective buyers to make a sales pitch, each call being independent of the others. Each call has a 15% chance of resulting in a sale. His daily goal is to make 3 sales, and he can make only 20 calls in a day. What is the probability that he does not meet his daily goal?

Hypergeometric distribution

- **Hypergeometric distribution** applies to experiments where a sample of n elements is selected without replacement from a population of N elements, which contains A successes. This distribution describes the probability of observing a specific number of successes in the sample.
- for $\max\{0, n - (N - A)\} \leq k \leq \min\{n, A\}$

$$P(X = k) = \frac{\binom{A}{k} \binom{N-A}{n-k}}{\binom{N}{n}}$$

- to show that $\sum_k P(X = k) = 1$ it suffices to compare coefficients in

$$(1+x)^N = (1+x)^A (1+x)^{N-A}$$

- $E(X) = n \frac{A}{N}$ and $Var X = n \frac{A}{N} \frac{N-A}{N} (1 - \frac{n-1}{N-1})$

Properties of hypergeometric distribution

Theorem 35: Let $N \rightarrow \infty$, $A \rightarrow \infty$ and $\frac{A}{N} \rightarrow p$, $0 < p < 1$. Then for every fixed n and $k = 0, 1, \dots, n$,

$$P(X = k) \xrightarrow{N \rightarrow \infty} \binom{n}{k} p^k (1 - p)^{n-k}.$$

- I.e., the hypergeometric distribution converges to the binomial distribution as the size of the finite population becomes large.

Theorem 36: Let X and Y be independent rv's with binomial distributions $X \sim \text{BIN}(m, p)$ and $Y \sim \text{BIN}(n, p)$. Then

$$P(X = k | X + Y = r) = \frac{\binom{m}{k} \binom{n}{r-k}}{\binom{m+n}{r}}.$$

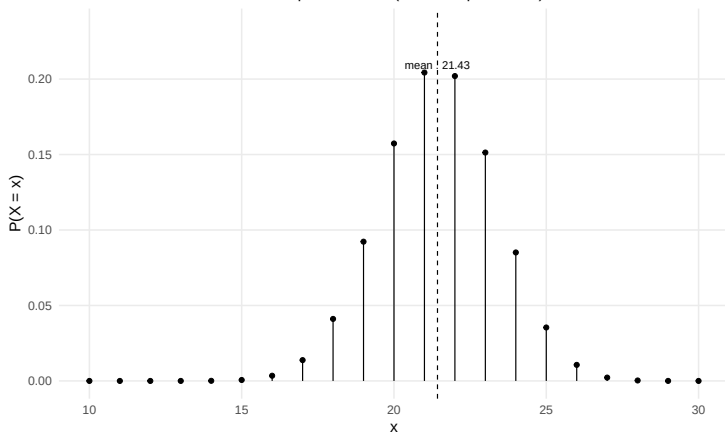
Hypergeometric distribution: Example

Example 66: Consider X with hypergeometric distribution with parameters $N = 70$, $A = 50$ and $n = 30$.

The graph of the **probability mass function**

Hypergeometric Distribution: $N = 70$, $A = 50$, $n = 30$

X = number of successes in a sample of size 30 (without replacement)



Hypergeometric distribution

Example 67: A class consists of 10 boys and 12 girls. The teacher randomly selects 6 children. Is it more likely that she selects 3 boys and 3 girls or that she selects 2 boys and 4 girls?

- A generalization: **Pólya urn scheme:**

The urn (population) contains N balls (elements), with A balls of one type and $N - A$ balls of another type. Each time a ball is sampled, it is returned to the urn, and additional c balls of the same type is added to the urn.

- If $c = 0$, this results in a binomial distribution.
- If $c = -1$, this results in a hypergeometric distribution.

Poisson distribution

- A random variable X is said to follow a **Poisson distribution** with parameter $\lambda > 0$, if

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, \dots$$

We denote this as $X \sim POI(\lambda)$

- The Poisson distribution can be considered a limiting case of the binomial distribution.

Theorem 37:

If $p \rightarrow 0, n \rightarrow \infty$ and $np \rightarrow \lambda > 0$ then

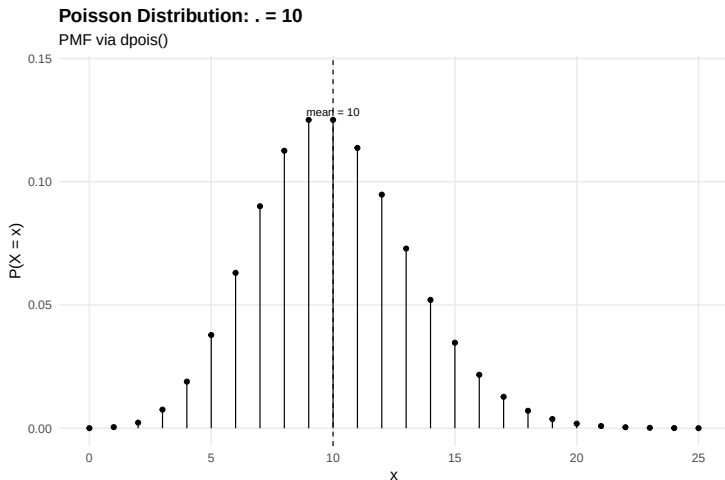
$$\lim \binom{n}{k} p^k (1-p)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}.$$

- $E(X) = \lambda$ and $Var X = \lambda$
- $m_X(t) = e^{\lambda(e^t - 1)}, \quad t \in \mathbb{R}$

Poisson distribution: Example

Example 68: Consider $X \sim POI(10)$.

The graph of the **probability mass function**



Properties of Poisson distribution

- The Poisson distribution is closed under the addition of independent random variables.

Theorem 38: If $X \sim POI(\lambda_1)$ and $Y \sim POI(\lambda_2)$ are independent, then

$$X + Y \sim POI(\lambda_1 + \lambda_2).$$

- If X and Y are independent random variables, both following Poisson distributions, then the conditional distribution of X given $X + Y$ is a binomial distribution.

Theorem 39: If $X \sim POI(\lambda_1)$ is independent of $Y \sim POI(\lambda_2)$ then for $k = 0, \dots, n$

$$P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}.$$

Application of Poisson distribution

- The number of occurrences of some event in a given time interval (such as customer arrivals at service stations; fire alarms, phone calls at call centers, etc.) can often be modelled using a Poisson distribution.
- Recall that we say f is $o(x)$ if $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$.
 - If $\lim_{x \rightarrow 0} \frac{h(x)}{x} = c$ then $h(x) = cx + o(x)$.
 - If $f_1, \dots, f_N$ are $o(x)$ then also $f_1 + \dots + f_N$ is $o(x)$.
- **Assumption 1:** The number of events occurring in two non-overlapping time intervals are independent.
- **Assumption 2:** The probability of at least one event occurring in an interval of length Δt is $\lambda \Delta t + o(\Delta t)$ for some constant $\lambda > 0$.
- **Assumption 3:** The probability of two or more events occurring in an interval of length Δt is $o(\Delta t)$.

Application of Poisson distribution

- Let $N[0, t)$ denote the number of events that occur before time t , and define $P_n(t) := P(N[0, t) = n)$, the probability of exactly n events prior time t . Then under assumptions 1, 2 and 3 we have

$$P_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, 2, \dots$$

Example 69: Suppose that traffic accidents at a given intersection occur according to a Poisson process. The rate on Saturdays is twice the rate on weekdays, and the rate on Sundays is twice the rate on Saturdays. The total rate for the week is about five accidents. What is the probability of exactly four accidents occurring on weekdays in a given week?

Uniform distribution

- We write $X \sim U[A, B]$, $A < B$
- Density function

$$f_X(x) = \begin{cases} \frac{1}{B-A}, & \text{if } A \leq x \leq B; \\ 0, & \text{otherwise.} \end{cases}$$

- Distribution function

$$F_X(t) = \begin{cases} 0, & \text{if } t < A; \\ \frac{t-A}{B-A}, & \text{if } A \leq t \leq B; \\ 1, & \text{if } t > B. \end{cases}$$

- $E(X) = \frac{A+B}{2}$, $Var X = \frac{(B-A)^2}{12}$
- Moment generating function

$$m_X(t) = \begin{cases} \frac{e^{Bt} - e^{At}}{t(B-A)}, & t \neq 0; \\ 1, & t = 0. \end{cases}$$

Exponential distribution

- **Exponential distribution** describes the time between events in a Poisson process. It is commonly used for modeling the random time until the occurrence of some event or length of occurrence, e.g. life expectancy of some device, modeling of response times of computer servers, etc.
- We write $X \sim EXP(\lambda)$, $\lambda > 0$.
- Cumulative distribution function

$$F_X(x) = \begin{cases} 0, & x < 0; \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

- $E(X) = \frac{1}{\lambda}$, $Var X = \frac{1}{\lambda^2}$
- Moment generating function

$$m_X(t) = \frac{\lambda}{\lambda - t} \text{ for } t < \lambda \text{ (else it is } +\infty).$$

Theorem 40: (memoryless property of $EXP(\lambda)$)

If X is a random variable with exponential distribution then for all $x, y > 0$

$$P(X > x + y | X > y) = P(X > x).$$

Memory-less property

Example 70: Studies of a single-machine-tool system showed that the mean time the machine operates before breaking down is 10 hours.

- i) Determine the failure rate and find the probability that the machine operates for at least 12 hours before breaking down
- ii) If the machine has been operating for 8 hours, what is the probability that it will last another 4 hours?

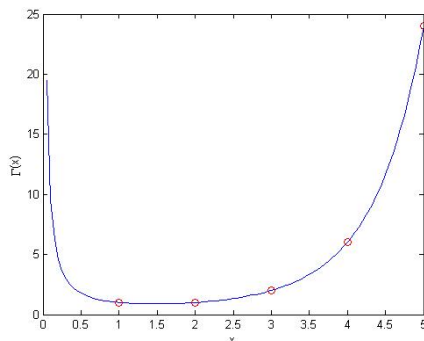
Gamma function

Definition 27: For $t \geq 0$ we define **gamma function** as

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} dx.$$

Properties:

- $\Gamma(1) = \Gamma(2) = 1$
- $\Gamma(t+1) = t\Gamma(t)$ for $t \in \mathbb{R}_+$
- $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$



Gamma function is a natural extension of factorial function to positive real numbers

Gamma distribution

- We write $X \sim GAM(\alpha, \lambda)$
- Density function

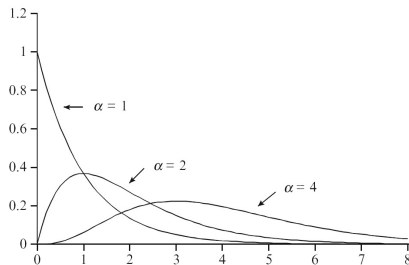
$$f(x) = \begin{cases} Cx^{\alpha-1}e^{-\lambda x}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

- α is the shape parameter and λ is the scale parameter
- Normalizing constant $C = \frac{\lambda^\alpha}{\Gamma(\alpha)}$

- $E(X) = \frac{\alpha}{\lambda}, Var X = \frac{\alpha}{\lambda^2}$
- Moment generating function

$$m_X(t) = \frac{1}{(1 - \frac{t}{\lambda})^\alpha} \quad \text{for } t < \lambda$$

- $m_k = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)\lambda^k}$



Gamma distributions, $\lambda = 1$.

Properties of Gamma distribution

Theorem 41: If $X \sim GAM(\alpha_1, \lambda)$ and $Y \sim GAM(\alpha_2, \lambda)$ are independent then $X + Y \sim GAM(\alpha_1 + \alpha_2, \lambda)$.

Theorem 42: If $X \sim GAM(\alpha, \lambda)$ then $Y = 2\lambda X \sim GAM(\alpha, \frac{1}{2})$.

Special cases:

- for $\alpha = 1$, $GAM(1, \lambda) = EXP(\lambda)$
- let $Y = Z^2$ and $Z \sim N(0, 1)$. Then

$$f_Y(y) = \begin{cases} \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} e^{-\frac{y}{2}}, & y \geq 0; \\ 0, & \text{otherwise,} \end{cases}$$

and hence $Y \sim GAM\left(\frac{1}{2}, \frac{1}{2}\right)$.

χ^2 distribution

Definition 28: For integer ν , the distribution $GAM\left(\frac{\nu}{2}, \frac{1}{2}\right)$ is called chi-square distribution with ν degrees of freedom.

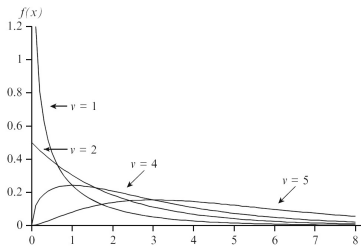
We write $X \sim \chi_\nu^2$.

- let $Y = Z^2$ and $Z \sim N(0, 1)$. Then $Y \sim \chi_1^2$.

Theorem 43:

- a) If $Z_1, \dots, Z_n$ are iid with $N(0, 1)$ then

$$X = Z_1^2 + \dots + Z_n^2 \sim \chi_n^2.$$



Chi-square distributions.

- b) If $X_1, \dots, X_n$ are independent rv's with $\chi_{\nu_1}^2, \dots, \chi_{\nu_n}^2$, respectively, then

$$Y = X_1 + \dots + X_n \sim \chi_\nu^2, \quad \text{with } \nu = \nu_1 + \dots + \nu_n.$$

- χ^2 distribution is very important for testing hypotheses

Normal distribution

- **Normal distribution** is the single most important distribution in probability theory and statistics. It is widely used in diverse areas of applications, e.g., modeling the heights and weights of human and animal populations.
- We write $X \sim N(\mu, \sigma^2)$
- Density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}.$$

- $E(X) = \mu, \text{Var}X = \sigma^2$
- Moment generating function

$$m_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}, \quad t \in \mathbb{R}.$$

Theorem 44: Let X and Y be independent rv's with $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$. Then $U = \alpha X + \beta Y$ has distribution $N(\alpha\mu_1 + \beta\mu_2, \alpha^2\sigma_1^2 + \beta^2\sigma_2^2)$.

Normal distribution

Example 71: Assume that the height of men in a certain population follows a normal distribution with mean $\mu_M = 175$ centimeters and standard deviation $\sigma_M = 5$ centimeters. The height of women is also normally distributed, with mean $\mu_W = 170$ centimeters and $\sigma_W = 4$ centimeters.

If one man and one woman are selected at random, what is the probability that the woman selected is taller than the man selected?

Example 71 cont.: Chances of $X \sim N(175, 25)$ being negative are of order $\Phi(-35) = 1 - \Phi(35) < 10^{-100}$.

Lognormal distribution

- **Lognormal distribution** is widely applicable in modeling, for example, where there is a multiplicative product of many small independent factors, e.g., the long-term return rate on a stock investment can be considered as the product of the daily return rates.

Definition 29: X has lognormal distribution if $Y = \log X$ has normal distribution.

- Density function

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}}, \quad x > 0.$$

- $E(X) = e^{\mu + \frac{\sigma^2}{2}}, \text{Var} X = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$

Example 72: If $Y \sim N(0, 1)$ and $Y = \log X$ then X has lognormal distribution with $E(X) = e^{\frac{1}{2}}$ and not $e^{E(Y)} = 1$!

Expectation of random vectors

- For a random vector X , we can extend the concept of expected value to a **vector of expected values**, and the concept of variance to a **variance matrix** of the random vector.

Definition 30: Let $X = (X_1, \dots, X_n)^\top$ be a random vector. Then

- a) the expected value of X is a vector

$$E(X) = (E(X_1), \dots, E(X_n))^\top$$

- b) if $E(X_i^2) < \infty, i = 1, \dots, n$, then $Var X$, the variance matrix of X , is defined as

$$Var X = \begin{pmatrix} Var X_1 & Cov(X_1, X_2) & \cdots & Cov(X_1, X_n) \\ Cov(X_1, X_2) & Var X_2 & \cdots & Cov(X_2, X_n) \\ \vdots & & \ddots & \vdots \\ Cov(X_1, X_n) & Cov(X_2, X_n) & \cdots & Var X_n \end{pmatrix} \\ = E((X - EX)(X - EX)^\top).$$

Variance matrix and correlation matrix

- c) if $E(X_i^2) < \infty, i = 1, \dots, n$, then $Corr X$, the correlation matrix of X , is defined as

$$Corr X = \begin{pmatrix} 1 & Corr(X_1, X_2) & \cdots & Corr(X_1, X_n) \\ Corr(X_1, X_2) & 1 & \cdots & Corr(X_2, X_n) \\ \vdots & & \ddots & \vdots \\ Corr(X_1, X_n) & Corr(X_2, X_n) & \cdots & 1 \end{pmatrix}$$

Theorem 45: (without proof) (Hölder inequality)

Let $E|X|^p < \infty$ and $E|Y|^q < \infty, p, q > 0, \frac{1}{p} + \frac{1}{q} = 1$. Then

$$E(XY) \leq (E|X|^p)^{\frac{1}{p}} (E|Y|^q)^{\frac{1}{q}}.$$

Special case: $E(XY) \leq \sqrt{E(X^2)E(Y^2)}$.

Properties of variance matrix

Theorem 46: (without proof)

Let $Var X_i < \infty, i = 1, \dots, n$. Then

- i) the variance matrix of X is symmetric and positive semidefinite;
- ii) for any $a = (a_1, \dots, a_n)^\top$ and $B = (b_{ij})_{m \times n}$

$$Var(a + BX) = B Var X B^\top;$$

- iii) if $X_1, \dots, X_n$ are mutually independent, or at least uncorrelated, then $Var X$ is a diagonal matrix;
- iv) if $Var X$ is singular then there exist nonzero real constants $a_1, \dots, a_{n+1}$ and $Q \subset S$ with $P(Q) = 1$ such that

$$\sum_{i=1}^n a_i X_i(s) = a_{n+1}$$

for all $s \in Q$.

Bivariate normal distribution

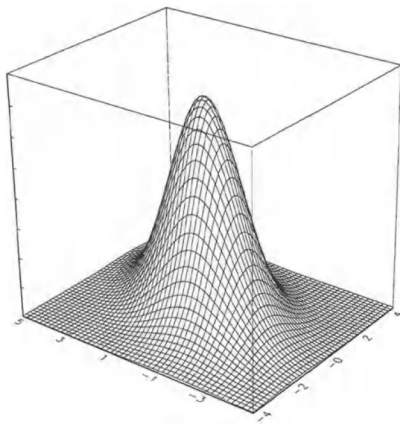
- $X \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right)$, where $\mu_1, \mu_2 \in \mathbb{R}, \sigma_1, \sigma_2 > 0, \rho \in [-1, 1]$.

- Joint density function

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \cdot \exp \left\{ -\frac{1}{2(1-\rho^2)} \left(\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{x-\mu_1}{\sigma_1}\frac{y-\mu_2}{\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right) \right\}$$

- Both X_1 and X_2 have marginal normal distribution $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively
- $E(X_1) = \mu_1, E(X_2) = \mu_2$
- $Var(X_1) = \sigma_1^2, Var(X_2) = \sigma_2^2, Corr(X_1, X_2) = \rho$

Bivariate normal distribution



Example 71 cont.: Assume the heights of siblings are jointly normally distributed and additionally correlated with $\rho = 0.6$. If we randomly sample a brother and a sister, what is the probability that the sister is taller than her brother?

Recommended reading (week 7)

For the next lecture Part 7, please, read

- sections 9.1, 9.2, 9.3, 9.4, 9.5, 9.6 from Bartoszynski and Niewiadomska-Bugaj (33 pages)
- sections 5.1, ... 5.8, from Mittelhammer (30 pages)